



Sportec Solutions

KPI - Documentation

Author: DFL / Sportec Solutions

Version: 0.0.3

Status: 19.02.2020



BUNDESLIGA

OFFIZIELLE
SPIELDATEN



BUNDESLIGA

OFFIZIELLE
SPIELDATEN

Document history

Version	Change	Date	Author
0.0.1	Document created	07.08.19	G. Anzer
0.0.2	Addition of selected columns	24.10.19	G. Anzer
0.0.3	Addition of xPass Values	19.02.20	G. Anzer

Table of contents

1	Introduction	4
2	Technical description.....	5

1 Introduction

The **Data Library** is an information system belonging to the DFL Deutsche Fußball Liga GmbH, which collects, stores, refines and distributes the **Official Match Data** for matches in the Bundesliga and Bundesliga 2, the Relegation Play-Offs and the Supercup.

The KPI Table described here provides automatically generated additional information and key performance indicators (KPI) to the raw event data from a match. It is based on the synchronized event- and tracking data of a match.

2 Technical description

The table contains all events that were manually collected throughout a match. Whenever possible the event is synchronized to the tracking data. Both for passes and shots, we provide additional information, which is computed based on the tracking data.

Information about passes include the exact frames they reached their target, the amount of pressure the passer and receiver were under amongst other things.

For shots we provide the corresponding Expected Goal (=xG) values and several features, that are used to predict the probability.

The column **Availability** in the following table describes for what type of events this attribute is computed.

Structure:

Column Name	Value	Description	Availability
GDCP_EVENT_ID	Integer	Unique ID of the event	All Events
EVENT_ID	Integer	Unique ID of the match event within the specified match	All Events
MUID	String	Unique ID of the match	All Events
MATCH_DAY	Integer	Match day of match	All Events
CUID1	String	Unique ID of the acting team	All Events
Club1	String	Name of the acting team	All Events
Club1_Three_Letter_Code	String	Three letter code of the acting team	All Events
PUID1	String	Unique ID of the acting player	All Events
Player1	String	Name of the acting player	All Events
SHIRT_NUMBER1		Shirt number of the acting player	All Events
PUID2	String	Unique ID of the secondary player (e.g. the receiver of a pass)	All Events
Player2	String	The name of the secondary player	All Events
SHIRT_NUMBER2		Shirt number of the secondary player	All Events
CUID2	String	Unique ID of the secondary player's team	All Events
Club2	String	Name of the secondary player's team	All Events
Club2_Three_Letter_Code	String	Three letter code of secondary player's team	All Events
GDCP_EVENT_TIME	Datetime	Time of the event according to the event data	All Events
TRACKING_TIME	Datetime	Estimated time of the event according to the tracking data	All Events
FRAME_NUMBER	Integer	Estimated tracking frame of the event	All Events
X_EVENT	[float]	X-Position of the event in meters according to the event data	All Events

DFL-03-EventData-Match-Raw Feed / Convenient Translation

Column Name	Value	Description	Availability
Y_EVENT	[float]	Y-Position of the event in meters according to the event data	All Events
X_TRACKING	[float]	Estimated X-Position of the event in meters according to the tracking data	All Events
Y_TRACKING	[float]	Estimated Y-Position of the event in meters according to the tracking data	All Events
SUBTYPE	String	Event subtype (e.g. "Pass", "BallClaiming" ...)	All Events
NOMATCH	Boolean	"TRUE" if our matching algorithm was unable to synchronize this event with the tracking data. "FALSE" otherwise.	All Events
Goingright	Boolean	Describes the attacking direction of the acting team. "TRUE" if the team is playing from the left to right (according to the x-axis in the tracking data). "FALSE" otherwise.	Passes
EVALUATION	String	Describes the success of a pass. If the player receiving the ball is on the same team as the player making the play, it is described as successful. If the player receiving the ball is from the opposing team, the play is unsuccessful. If no one received the ball from the play but the team remains in possession, it is described as successful. The following characteristics are possible: successfullyCompleted successful unsuccessful	Passes
PRESSURE	[float]	Describes the amount of pressure the player passing the ball is under, at the time of the pass. This value is calculated by logarithmically adding the pressure values of all opposing players. The individual pressure levels are calculated based on the positioning, the playing direction, the speed and the movement of the players.	Passes
PRESSER	String	Unique ID of the player exerting the highest amount of pressure on the passer	Passes
Presser_Name	String	Name of the player exerting the highest amount of pressure on the passer	Passes
NORECEIVER	Boolean	"TRUE" if the pass was unsuccessful or our matching algorithm was unable to identify the frame in which the pass was received. "FALSE" otherwise.	Passes

DFL-03-EventData-Match-Raw Feed / Convenient Translation

Column Name	Value	Description	Availability
RECFRM	Integer	Estimated tracking frame of the pass reception.	Passes
XRec	[float]	Estimated X-Position of the pass reception in meters according to the tracking data	Passes
YRec	[float]	Estimated Y-Position of the pass reception in meters according to the tracking data	Passes
PRESSURE_RECEIVER	[float]	Describes the amount of pressure the player receiving the pass is under, at the time of the pass reception. This value is calculated by logarithmically adding the pressure values of all opposing players. The individual pressure levels are calculated based on the positioning, the playing direction, the speed and the movement of the players.	Passes
PRESSER_REC	String	Unique ID of the player exerting the highest amount of pressure on the player receiving the pass	Passes
Presser_Rec_Name	String	Name of the player exerting the highest amount of pressure on the player receiving the pass	Passes
Bypassed_Players	Integer	Bypassed players is defined as the positive difference between the number of players that are closer to their own goal than the ball after and prior of a pass.	Passes
Bypassed_Defenders	Integer	Bypassed defenders is defined as the positive difference between the number of defenders (= the five players closest to their own goal at the time of the pass) that are closer to their own goal than the ball after and prior to a pass.	Passes
xG	[float]	The "Expected Goal" (xG) value of a shot. This value denotes the a priori probability of a shot being converted to a goal. Hence its value ranges from [0,1]. The probability is estimated using both tracking and event data and applying a machine learning model, that was trained on more than 100.000 shots.	Shots
PENALTY	Boolean	"TRUE" if the shot was a penalty. "FALSE" otherwise.	Shots
AssistTyp	String	Type of Assist, leading up to the shot.	Shots
DISTANCE	[float]	Distance in meters between the location of the shot and the center of the opposing goal.	Shots

DFL-03-EventData-Match-Raw Feed / Convenient Translation

Column Name	Value	Description	Availability
ANGLE	[float]	Angle in radians between the location of the shot and the two posts of the opposing goal.	Shots
FREEKICK	Boolean	"TRUE" if the shot taken was a direct free kick, "FALSE" otherwise.	Shots
FOOT	Boolean	"TRUE" if the shot was taken with a foot, "FALSE" otherwise.	Shots
SPEED	[float]	The speed in [km/h] the player attempting the shot was travelling at the time of the shot.	Shots
PRESSURE_ON_SHOT	[float]	The amount of pressure the player attempting the shot was under at the time of the shot.	Shots
DEFENDERS	Integer	The number of defenders in the line of the shot.	Shots
DISTANCE_GOALKEEPER PER	[float]	The distance in meters the goalkeeper between the goalkeeper and the center of the goal at the time of the shot.	Shots
GK_INGOAL	Boolean	"TRUE" if the goalkeeper is in the line of the shot. "FALSE" otherwise.	Shots
xPass	[float]	The "Expected Pass" (xPass) value of a pass. It denotes the a priori probability of a pass being successfully completed to its intended target. Hence its value ranges from [0,1]. The probability is estimated using both tracking and event data and applying a machine learning model, that was trained on more than 1000.000 passes.	Passes
PossSpeed	[float]	The possible speed range with which the pass would theoretically likely still be successful according to our ballistic model. Its value denotes the width of the window expressed as a percentage compared to the original pass speed.	Passes
PossAng	[float]	The possible angle range with which the pass would theoretically likely still be successful according to our ballistic model. Its value denotes the width of the window expressed as an angle in radians.	Passes
Dis_nearest_opp_rec	[float]	The distance [m] of the nearest opposing player to the intended receiver of a pass.	Passes
Dist_of_pass	[float]	The distance [m] between the passer and at the intended receiver at the time of the pass.	Passes

DFL-03-EventData-Match-Raw Feed / Convenient Translation

Column Name	Value	Description	Availability
Nearest_Defender_to_passing_lane	[float]	The distance [m] of the closest opposing player to the direct path between the passer and the intended pass receiver.	Passes
Angle_of_pass	[float]	The angle [in radians] of the pass relative to the playing direction.	Passes
OneTouch	Integer	"1" if the player performed the pass with his first contact. "0" otherwise.	Passes
Time_of_Possesion	[float]	The time [s] a player had ball possession before the pass was played.	Passes
BlockProb	[float]	The probability [0,1] that a pass will be blocked within the first 5 meters. This probability is already incorporated in the xPass model.	Passes
Opp_Players_in_path	Integer	The number of opposing players in the line of the pass (+- 5 meters to both sides).	Passes